Srikanth Mangipudi

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GitHub | Linkedin

Professional Experience

Quantum AI Global Sep 2024 – Present
Software Development Engineer Hyderabad

- Independently owned a Next.js admin portal: **17 pages and 61 API routes** for account provisioning, subscriptions, orders, finance, and audit logs; shared validation across UI/APIs and prevented conflicting edits with optimistic concurrency.
- Rebuilt the **customer dashboard** with parallel module loading and browser navigation support; unified sign-in across three applications using coordinated server-side token refresh and trusted-origin restrictions.
- Built authentication, authorization and software licensing services using **FastAPI** and **Keycloak**; secured login/password-reset flows across 5 identity realms and enforced role-based access and subscription-based feature limits.
- Designed **PostgreSQL** product configuration for 10+ software and 5 hardware-only products using JSONB columns; delivered 38+ schema migrations, audit trails, and 257 automated tests.
- Integrated **Razorpay** checkout/autopay and activation-based billing; fixed license renewals, seat counts, and invoices, resolved 60–70 frontend defects, and authored **CI/CD** workflows and deployment documentation.
- Built a **React/TypeScript** network simulator with Cytoscape.js, validation, and undo/redo; visualized simulation results streamed from Python services over gRPC.
- Independently delivered 11 **Python/Qt** modules, separating threaded acquisition from visualization for responsive interfaces; supported offline simulation, statistical validation, and Windows releases.
- Engineered **Electron.js/React.js** application for device control with USB discovery, serialized command queues, timeout recovery, and device-state synchronization after reconnection.

Vehant Technologies Jul 2023 – Dec 2023
Software Development Engineer Intern Noida

- Developed a **C++** camera-control service using ONVIF/SOAP and WS-Security to convert scene coordinates into authenticated pan, tilt, and zoom commands.
- Built **Kafka**-based asynchronous command and device-status workflows with **multithreaded** controllers, thread-safe queues, priority scheduling, and retries.

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL
Frontend: React, Next.js
Backend / APIs: FastAPI, Express.js, REST APIs, gRPC
Databases: PostgreSQL, MySQL
Identity / Security: Keycloak, OAuth2 / OIDC, RBAC
Deployment: Linux, Nginx, Docker, GitHub Actions (CI/CD)
Testing / Version Control: Pytest, Git

Projects

Pantrix – Online Grocery Store | Java, Spring Boot, MySQL, Docker Live | Code

- Built customer accounts, product search, cart/checkout, invoices, and admin workflows using **JPA/Hibernate and JSP/Bootstrap**; deployed with Docker on Render and cloud MySQL.

TomiRen – E-commerce Platform | TypeScript, React, Express, PostgreSQL Code

- Contributed product APIs, PostgreSQL search and suggestions, shared validation, and filters; evaluated query execution plans on a **50k-product test dataset**.

Education

BITS Pilani, Hyderabad Campus Nov 2020 – Aug 2024
B.E. Electronics and Instrumentation Engineering; Minor in Computing and Intelligence
Publication: Co-author, *Laser Frequency Locking for Quantum Sensing Applications Using AI/ML Techniques*, IEEE COMSNETS 2025.